

Advanced (Carolina Reaper)

Q1. Convert $x^2 + 6x + 8$ to vertex form.

- (A) $(x + 3)^2 - 1$
 (B) $(x + 3)^2 + 1$
 (C) $(x + 4)^2$
 (D) $(x + 2)^2 + 4$
 (E) $(x + 4)^2 - 8$

Q2. Complete the square for $x^2 + 4x$.

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x + 4$
 (C) $x^2 + 4x - 4$
 (D) $(x + 2)^2$
 (E) $(x - 2)^2$

Q3. What is the vertex of $y = (x - 2)^2 + 3$?

- (A) (2, 0)
 (B) (0, 3)
 (C) (2, 3)
 (D) (-2, 3)
 (E) (2, -3)

Q4. Convert $x^2 - 8x + 12$ to vertex form.

- (A) $(x - 4)^2 - 4$
 (B) $(x - 2)^2 - 8$
 (C) $(x + 4)^2 - 4$
 (D) $(x + 2)^2 - 8$
 (E) $(x - 4)^2$

Q5. What is the equation of the parabola with vertex (1, 2) and y -intercept (0, 3)?

- (A) $y = (x - 1)^2 + 2$
 (B) $y = (x + 1)^2 - 2$
 (C) $y = (x - 1)^2 - 1$
 (D) $y = x^2 + 2x + 1$
 (E) $y = x^2 - 2x + 3$

Q6. Complete the square for $x^2 - 6x$.

- (A) $x^2 - 6x + 9$
 (B) $x^2 + 6x + 9$
 (C) $x^2 - 6x - 9$
 (D) $(x - 3)^2$
 (E) $(x + 3)^2$

Q7. What is the vertex of $y = x^2 + 2x - 3$ after completing the square?

- (A) (-1, -4)
 (B) (-1, 2)
 (C) (-1, -2)
 (D) (1, -2)
 (E) (-1, 4)

Q8. Convert $y = x^2 - 4x - 3$ to vertex form.

- (A) $y = (x - 2)^2 - 7$
 (B) $y = (x + 2)^2 + 7$
 (C) $y = (x - 2)^2 + 1$
 (D) $y = (x + 2)^2 - 7$
 (E) $y = x^2$

Open Ended Questions (FRQ)**Q9.** Complete the square and find the vertex of $y = x^2 + 8x + 10$. Show all steps and express your final answer in vertex form.**Q10.** Given $y = ax^2 + bx + c$, describe the process for completing the square and express y in vertex form ($y = a(x - h)^2 + k$). Use an example to support your explanation.**Chef's Tips**

- Read the questions carefully
- Check your work
- Use the space provided